

NGUYEN DUY DUC

AI ENGINEER

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Hanoi, Vietnam

SUMMARY

Final-year Computer Science student focused on applied AI and production-oriented LLM systems. Experienced in retrieval evaluation, legal-aware ingestion, AI observability, backend integration, and cloud deployment. Built AI systems that are measurable, traceable, and ready for real workflows.

TECHNICAL SKILLS

- Applied AI:** LLM applications, RAG, LangGraph, hybrid retrieval, BM25, dense search, RRF, ChromaDB, BGE-M3, evaluation
- Machine Learning:** Python, PyTorch, preprocessing, training pipelines, experiment tracking, reinforcement learning, world models
- Backend & APIs:** FastAPI, Spring Boot, REST APIs, JWT, Auth/RBAC, SQLAlchemy, JDBC/JPA
- Data & Cloud:** PostgreSQL, MySQL, Redis, Docker, Supabase, Render, Vercel, Hugging Face Spaces
- Frontend:** React, Vite, Tailwind CSS, React Router, Redux Toolkit Query
- Tools:** Git, GitHub, Jira, Postman, Swagger, VS Code, IntelliJ IDEA

WORK EXPERIENCE

AI Engineering Trainee

VinUni | May 2026 – Expected: August 2026

- Focus: Applied AI Engineering, RAG Applications, Cloud Deployment, ML Infrastructure
- Project: Legolas AI | Legal Assistant MVP | 2-person team | Demo: <https://legolas-ai-staging.vercel.app>
 - Co-developed three end-to-end legal AI workflows: citation-grounded Q&A, template-based drafting, and document review with risk findings and source traceability.
 - Co-designed legal-aware hybrid ingestion, reducing chunks containing multiple article markers from 54.35% to 17.31% while preserving front matter and legal hierarchy.
 - Built and ran a 30-case multi-source benchmark, improving citation hit rate 46.67%→80.00%, legal-number recall 61.67%→90.00%, and multi-source fact score 60.00%→90.00% over the overlap baseline.
 - Deployed a split cloud architecture across Vercel, Render, Supabase/PostgreSQL, and Hugging Face Spaces; used Langfuse traces to investigate latency, provider fallback, and citation failures.

Fullstack Developer Intern – Sapo

Hanoi | Feb 2026 – May 2026

- Built RESTful product/category/warehouse APIs using Spring Boot, JWT authentication, validation, pagination, and transaction handling.
- Developed React modules and integrated frontend flows with backend APIs using fetch/axios, React Router, and Redux Toolkit Query.
- Practiced production-oriented software foundations including SQL optimization, web security basics, and AI-assisted development workflows.

QA Tester Intern – Shoop City

Remote – France | Jun 2024 – Aug 2024

- Executed functional and regression test cases, documented defects in Jira, and collaborated with developers to verify fixes.

PROJECTS

HanoiWorld – JEPA-based World Model for Autonomous Driving

(Group Project) | Python, PyTorch, DreamerV3, JEPA/V-JEPA

Built a driving world model for autonomous vehicle simulation, focusing on representation-level prediction rather than pixel-level reconstruction.

Contribution:

- Implemented JEPA-style observation encoding modules for learning compact driving-scene representations used by RSSM latent dynamics and actor-critic control.
- Contributed to architecture planning, dataset preprocessing, training pipeline setup, model training logic, and experiments across highway, merge, and roundabout simulations.
- Evaluated driving behavior using collision rate, episode reward, and scenario-level planning metrics.

GitHub: <https://github.com/CS-3331-Fundamental-of-AI/WorldModel-Self-Driving-Car>

EDUCATION

Troy University
B.S. Computer Science
2021–2026 | GPA: 3.1/4.0

Hanoi University of Science and Technology
Bachelor of Computer Science
2021 – 2026

CERTIFICATIONS

IELTS 7.0
MOS (Microsoft Office Specialist – Word)